CHAPTER 5

CENTRIPETAL FORCES

NEWTON'S first law of motion states that every body will continue in a state of rest or of uniform motion in a straight line unless acted upon by an external force. If a stone is fastened to the end of a piece of string and whirled around in a circular path, an inward pull must be continuously exerted to keep it travelling in a circle, the stone itself is exerting an outward radial pull trying to get away. If the speed is increased the pull becomes greater until the string snaps, the stone then flies off in a *straight line* at a tangent to the circle.

The outward radial force created by a body travelling in a circular path due to its natural tendency to fly off and travel in a straight line is termed the *centrifugal force*. The inward pull applied to counteract this and keep it on its circular path is termed the *centripetal force*, it is equal in magnitude to the centrifugal force and opposite in direction (Newton's third law of motion).

Consider a body moving at a constant speed of v around a circle of radius r. Referring to Fig. 64, at the instant it is passing point a its instantaneous velocity is v in the direction tangential to the circle at a; a little further around it is passing point b and its velocity is now v tangential to the circle at b. Although the speed is constant, the velocity has changed because there is a change of direction.

Let the movement from a to b be through a small angle θ. To find the change of velocity the vector diagram of velocities is drawn. If θ is considered a small angle then the base angles of the triangle are almost 90° and the change of velocity can be taken as $v \sin\theta$, $v \tan\theta$ or $v\theta$. For small angles:

$$\sin\theta = \tan\theta = \theta \text{ rad}$$

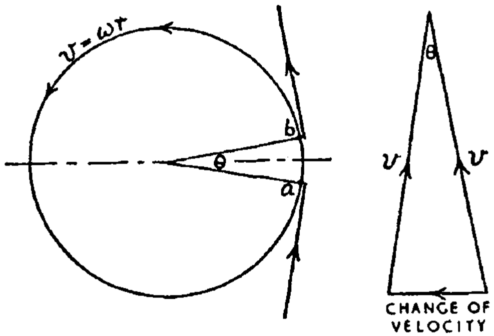

Fig. 64

Change in velocity $= v\theta$

If there is a change of velocity there is acceleration.

Time for body to move from a to b

$$= \frac{\text{distance}}{\text{speed}} = \frac{r\theta}{v}$$

$$\begin{aligned}\text{Acceleration} &= \text{change of velocity} \div \text{time} \\ &= v\theta \div \frac{r\theta}{v} \\ &= \frac{v\theta \times v}{r\theta} \\ &= \frac{v^2}{r}\end{aligned}$$

This is a special form of acceleration. The body is travelling at a constant speed, the acceleration is due to the constantly changing velocity (by changing direction) as the body travels around its circular path, and the acceleration is directed towards the centre of the circle. It is therefore distinguished from the more common cases by referring to it as *centripetal acceleration.*

Being circular motion, the velocity of the body may be more conveniently expressed in angular measurement ω rad/s instead of in linear units of v. The relation between these is $v = \omega r$.

Substituting this value of v:

$$\frac{v^2}{r} = \frac{\omega^2 r^2}{r} = \omega^2 r$$

$$\therefore \text{ Centripetal acceleration } = \frac{v^2}{r} \text{ or } \omega^2 r$$

Example. A ventilation fan, 0·5 m diameter rotates at 200 rev/min. Calculate the centripetal acceleration of the fan blade tips.

$$\omega = \frac{2\pi \times 200}{60}$$
$$= 20{\cdot}94 \text{ rad/s}$$
$$\text{Centripetal accln} = \omega^2 r$$
$$= 20{\cdot}94^2 \times 0{\cdot}25$$
$$= 109{\cdot}6 \text{ m/s}^2 \text{ (acting radially inward). Ans.}$$

The force required to produce the centripetal acceleration is called the centripetal force.

$$F = ma$$
$$\text{Centripetal force} = m\omega^2 r \text{ or } \frac{mv^2}{r}$$

This centripetal force is applied radially inwards to constrain a body to follow a circular or curved path. Centripetal force counterbalances the centrifugal force which acts radially outwards, equal in magnitude to the centripetal force, but in opposite in direction.

$$\text{Centrifugal force} = m\omega^2 r \text{ or } \frac{mv^2}{r}$$

Example. Calculate the pull on the root of a turbine blade due to centrifugal force when the rotor is rotating at 3000 rev/min, the mass of the blade is 0·2 kg and the radius from the axis of the rotor to the centre of gravity of the blade is 460 mm.

$$\frac{300 \text{ rev/min} \times 2\pi}{60} = 100\pi \text{ rad/s}$$
$$\text{Centrifugal force} = m\omega^2 r$$
$$= 0{\cdot}2 \times (100\pi)^2 \times 0{\cdot}46$$
$$= 9080 \text{ N} = 9{\cdot}08 \text{ kN} \qquad \text{Ans.}$$

Example. A body of 2 kg mass is attached to the end of a cord and swung around in a vertical plane of 0·75 m radius at a speed of 100 rev/min. Find (i) the centrifugal force set up, (ii) the tension in the cord when passing (*a*) top centre, (*b*) bottom centre, (*c*) a point 30° from bottom centre, (*d*) a point 60° from top centre.

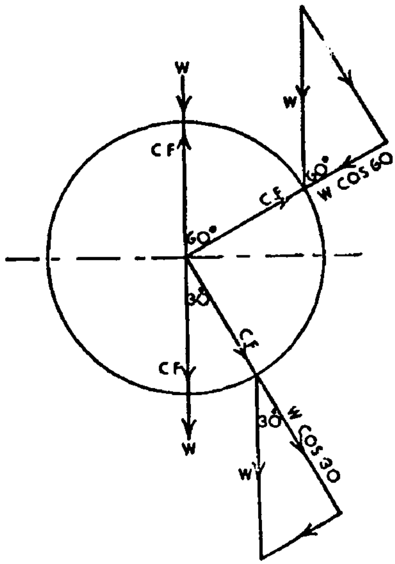

Fig 65

$$\frac{100 \text{ rev/min} \times 2\pi}{60} = 10{\cdot}47 \text{ rad/s}$$

Centrifugal force $= m\omega^2 r$

$= 2 \times 10{\cdot}47^2 \times 0{\cdot}75$

$= 164{\cdot}4$ N   Ans. (i)

Force of gravity on 2 kg mass (weight)

$= 2 \times 9{\cdot}81 = 19{\cdot}62$ N

When passing top centre, centrifugal force pulls directly upwards (always radially outwards) and the force of gravity acts vertically downwards, therefore,

Tension in cord = centrifugal force − weight

$= 164{\cdot}4 - 19{\cdot}62$

$= 144{\cdot}78$ N   Ans. (iia)

When passing bottom centre, centrifugal force pulls directly downwards, the force of gravity also acts downwards, therefore,

Tension in cord = centrifugal force + weight

$= 164{\cdot}4 + 19{\cdot}62$

$= 184{\cdot}02$ N   Ans. (iib)

When passing the point 30° to the bottom centre, the effect of the weight of the mass on the cord is its component in the direction of the cord, see Fig. 65, this is $W \cos 30°$ and acts in the same direction as the centrifugal force, therefore,

$$\begin{aligned}\text{Tension in cord} &= \text{centrifugal force} + W\cos 30° \\ &= 164{\cdot}4 + 19{\cdot}62 \times 0{\cdot}866 \\ &= 181{\cdot}39\ \text{N} \qquad\qquad \text{Ans. (iic)}\end{aligned}$$

At 60° from top centre the component of the weight is $W \cos 60°$ in the direction of the cord, but is in the opposite direction in which the centrifugal force acts, hence,

$$\begin{aligned}\text{Tension in cord} &= \text{centrifugal force} - W\cos 60° \\ &= 164{\cdot}4 - 19{\cdot}62 \times 0{\cdot}5 \\ &= 154{\cdot}59\ \text{N} \qquad\qquad \text{Ans. (iid)}\end{aligned}$$

SIDE SKIDDING AND OVERTURNING OF VEHICLES

When a car or lorry takes a bend in the road, around a corner or road island, it moves around a circular path and centrifugal force is created which tends to cause the vehicle to skid or overturn.

SIDE SKID. Treaded rubber tyres provide a good grip on the ground and, within limits, prevent broadside skidding. To slide a vehicle sideways, the force required to overcome friction depends upon the *coefficient of friction* between the tyres and the ground, the coefficient of friction is the ratio between the force required to overcome friction and the normal force between the surfaces. Thus if a vehicle weighing W stands on a horizontal ground (Fig. 66) the force between the surfaces is simply the weight W. If the coefficient of friction is represented by μ, the force to drag it sideways is $\mu \times W$ (see Chapter 6).

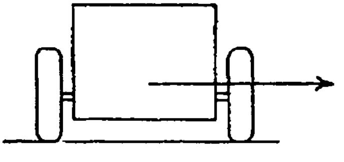

Fig. 66

For example, if the mass of a car is 1000 kg and the coefficient of friction between the tyres and ground is 0·6, the force to overcome friction is $0{\cdot}6 \times 100 \times 9{\cdot}81 = 5886$ N. This force could be reached and exceeded by the centrifugal force created when the car travels around a bend at an excessive speed.

Example. Calculate the speed at which a car will begin to skid sideways when turning a bend of 25 m radius, taking the coefficient of friction between tyres and ground as 0·7. If the coefficient of friction is halved due to worn tyres and wet road, what will now be the danger speed?

Let m = mass of car, in kg

then force between surfaces $= m \times 9{\cdot}81$ N

Force to overcome friction to push the car sideways $= 0{\cdot}7 \times m \times 9{\cdot}81$ N, and when the centrifugal force reaches this value skidding will occur.

$$\frac{mv^2}{r} = 0{\cdot}7 \times m \times 9{\cdot}81$$

$$v = \sqrt{0{\cdot}7 \times 9{\cdot}81 \times 25}$$

$$= 13{\cdot}1 \text{ m/s}$$

$$13{\cdot}1 \times 10^{-3} \times 3600 = 47{\cdot}17 \text{ km/h} \qquad \text{Ans. (i)}$$

Velocity varies directly as the square root of the coefficient of friction. If the coefficient of friction is halved, then,

$$\text{New danger speed} = 47{\cdot}17 \times \sqrt{0{\cdot}5}$$

$$= 33{\cdot}35 \text{ km/h} \qquad \text{Ans. (ii)}$$

OVERTURNING. If friction provides sufficient grip to prevent side-skid, there is the other danger of overturning when turning corners at excessive speeds, especially with vehicles which have a high centre of gravity. Referring to Fig. 67, the vehicle tilts about the point of contact of the ground and those wheels which are furthest from the centre of the circular path, marked o. The centrifugal force acts horizontally through the centre of gravity and the force of gravity acts vertically downwards through this point. If the centre of gravity is h above the ground, this is the leverage or perpendicular distance from the line of action of the centrifugal force to point o, the *overturning moment* is therefore $C.F. \times h$. The moment maintaining the vehicle on its four wheels is the weight multiplied by its perpendicular distance, this distance being half the track of the wheels. When the overturning moment becomes equal to the stabilising moment, overturning will commence (see Chapter 7).

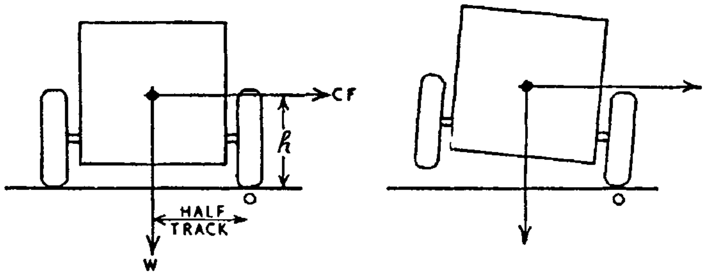

Fig. 67

Example. The track between the wheels of a lorry is 1·52 m and its centre of gravity is 0·9 m above the ground. Find the speed at which it will overturn when travelling around a bend of 25 m radius.

$$\text{Overturning moment} = \text{Stabilising moment}$$

$$\text{Centrifugal force} \times h = \text{Weight} \times \tfrac{1}{2}\,\text{track}$$

$$\frac{mv^2}{h} \times h = mg \times \tfrac{1}{2}\,\text{track}$$

$$v = \sqrt{\frac{9{\cdot}81 \times 0{\cdot}76 \times 25}{0{\cdot}9}}$$

$$= 14{\cdot}39 \text{ m/s}$$

$$14{\cdot}39 \times 10^{-3} \times 3600 = 51{\cdot}9 \text{ km/h} \qquad \text{Ans.}$$

BANKED TRACKS. Bends of roadways and railways are often banked, the incline to the horizontal being calculated on the average speed of vehicles passing around the bends. By inclining the road, the reaction of the ground which is normal (*i.e.* at right angles) to the surface, is inclined, and its two components are (i) an upward force to support the downward weight of the vehicle, and (ii) a horizontal force to counteract the centrifugal force of the vehicle. This is illustrated in Fig. 68.

The magnitude of the horizontal component to counteract the centrifugal force depends upon the tangent of the incline to which the road is banked and hence can be designed to completely balance the centrifugal force at any pre-determined speed.

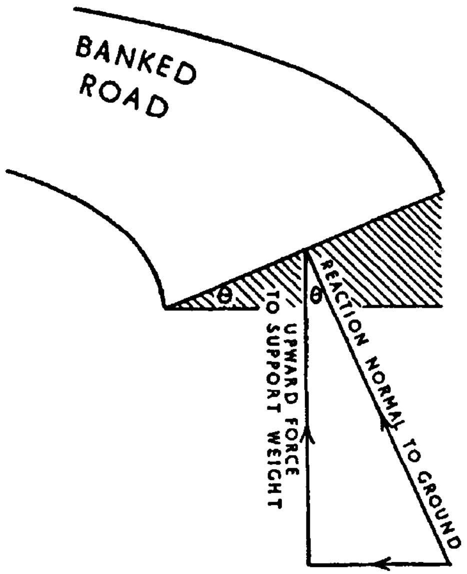

Fig. 68

Example. Find the angle to the horizontal to which a bend in the road should be banked so that there will be no tendency to skid when a vehicle is travelling at a speed of 72 km/h, the radius of the bend being 180 m. Neglect friction between tyres and ground.

$$\frac{72 \text{ km/h} \times 10^3}{3600} = 20 \text{ m/s}$$

Upward force to support weight of mass of m kg

$$= m \times 9{\cdot}81 \text{ N}$$

$$\text{tan of angle of incline} = \frac{\text{centrifugal force}}{\text{weight}}$$

$$= \frac{mv^2}{r \times 9{\cdot}81\, m}$$

$$= \frac{20^2}{180 \times 9{\cdot}81} = 0{\cdot}2266$$

Angle of incline = $12° \, 46'$ Ans.

In railway lines the same effect is obtained by raising the outer rail, in this case it is done to prevent an excessive side thrust on the flanges of the wheels.

Example. Calculate the height of the outer rail above the level of the inner rail around a curvature of 800 m radius of a railway of 1435 mm track so that there will be no side thrust on the wheel flanges when the train is travelling at 80 km/h.

Let θ = angle of inclination of outer rail above inner rail

h = super-elevation of outer rail

$$\frac{80 \text{ km/h} \times 10^3}{3600} = 22{\cdot}22 \text{ m/s}$$

$$\tan\theta = \frac{\text{centrifugal force}}{\text{weight}} = \frac{mv^2}{r \times mg}$$

$$= \frac{22{\cdot}22^2}{800 \times 9{\cdot}81} = 0{\cdot}06292$$

$$\theta = 3°\,36'$$

$$h = 1435 \times \sin 3°\,36'$$

$$= 1435 \times 0{\cdot}0628$$

$$= 90{\cdot}12 \text{ mm} \quad \text{Ans.}$$

BALANCING

If the centre of gravity of a rotating piece of machinery does not coincide with the centre of rotation, centrifugal force is set up which causes vibration as well as putting an extra load on bearings. To balance an eccentric load in its own plane place a counter-balance diametrically opposite, of such mass and radius from the centre so that it will create a centrifugal force of equal magnitude and opposite in direction to that of the eccentric load.

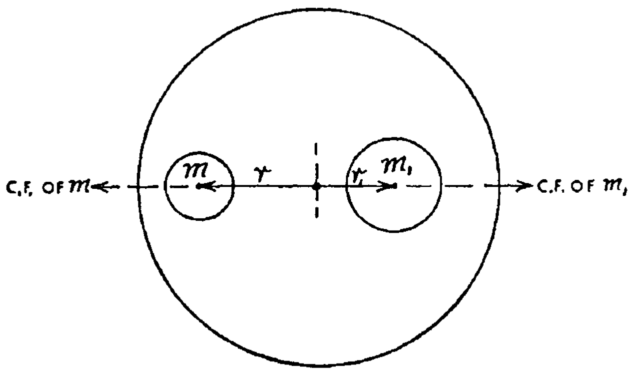

Fig. 69

Referring to Fig. 69, let m_1 be the mass to be balanced, and let r_1 be the distance from centre of rotation to the centre of gravity of the mass. Let m be the mass of the counter-balance placed diametrically opposite at a radius r. For balance, the centrifugal force of m must be equal to the centrifugal force of m_1 thus,

$$m\,\omega^2\,r = m_1\,\omega^2\,r_1$$

Since they both rotate at the same velocity, ω^2 cancels, therefore,

$$m \times r = m_1 \times r_1$$

Note that if the moments are taken about the centre of rotation, the turning moment of one mass is equal to the turning moment of the other and therefore the system is statically balanced (see Chapter 7).

The centre of gravity of the above masses must lie in the same plane (see side elevation in Fig. 70), otherwise an unbalanced couple will be set up when the system rotates, which will cause a rocking action. Taking moments about the line of action of one force, the unbalanced moment (couple) is *C.F.* × arm (or leverage) between the two forces (see Chapter 7).

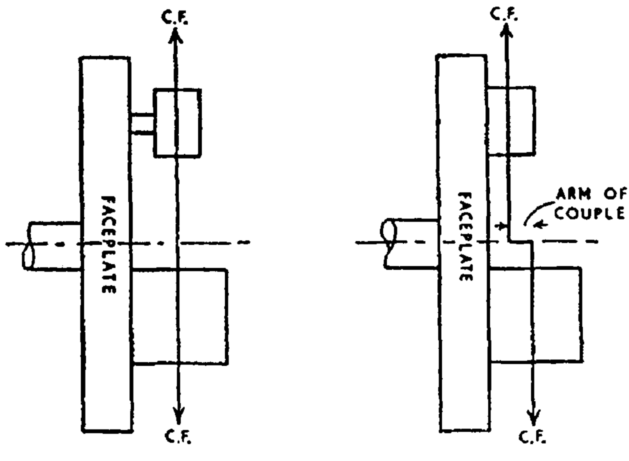

Fig. 70

Example. A valve-box casting of mass 27 kg is secured to the faceplate of a lathe in such a position that its centre of gravity is 90 mm radially from the centre of rotation and 114 mm axially from the surface of the faceplate. Calculate the radius at which to fix a counter-balance of 12·75 kg. If the centre of gravity of this counter-balance is 76 mm axially from the faceplate surface, calculate the unbalanced couple when rotating at 200 rev/min.

$C.F.$ of counter-balance = $C.F.$ of casting

$$m_1\,\omega^2\,r_1 = m_2\,\omega^2\,r_2$$

$$m_1 \times r_1 = m_2 \times r_2$$

$$12{\cdot}75 \times r_1 = 27 \times 90$$

$$r_1 = 190{\cdot}5 \text{ mm} \qquad \text{Ans.}$$

Thus, counter-balance should be fixed at a radius of 190·5 mm diametrically opposite the direction in which the centre of gravity of the casting lies.

$$\frac{200 \text{ rev/min} \times 2\pi}{60} = 20{\cdot}94 \text{ rad/s}$$

Unbalanced couple = C.F. of casting (or counter-balance) × Arm of couple

$$= m\,\omega^2\,r \times \text{arm}$$

$$= 27 \times 20{\cdot}94^2 \times 0{\cdot}09 \times (0{\cdot}114 - 0{\cdot}076)$$

$$= 40{\cdot}48 \text{ N m} \qquad \text{Ans.}$$

BALANCING A NUMBER OF ROTATING MASSES

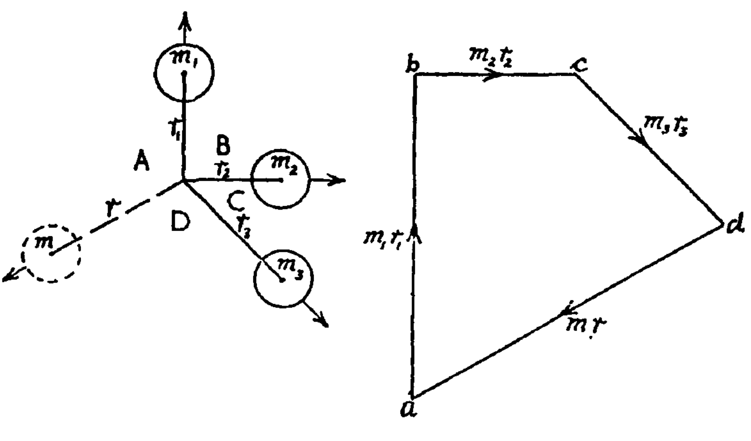

Fig. 71

If the vector diagram representing a number of coplanar forces meeting at a point forms a closed figure, the system is in equilibrium.

Centrifugal forces can be dealt with in the same manner. A number of rotating masses can be balanced if they are all in the same plane either by displacing them at such angular positions that their vector diagram forms a closed figure, or, if the positions of the existing masses cannot be arranged in this way, an additional mass can be included in such a position that the vector of its centrifugal force will close the vector diagram (see Fig. 71).

Example. Two masses are fixed at right angles to each other on a disc which is to rotate, one is 3 kg at a radius of 125 mm from the centre of rotation, and the other is 4 kg at 150 mm radius. Find the position to fix a balance mass of 7 kg to equalise the centrifugal forces.

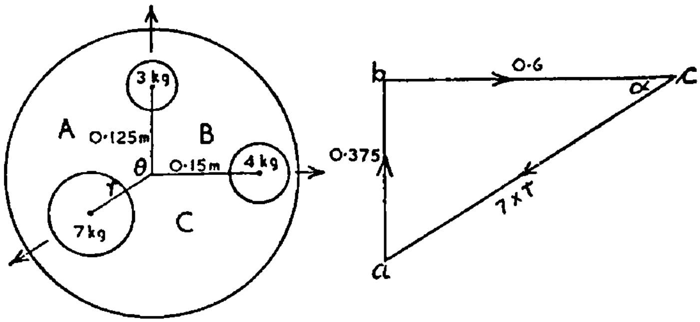

Fig. 72

Vector ab is drawn to represent force AB,

represented by $3 \times 0{\cdot}125 = 0{\cdot}375$

Vector bc is drawn to represent force BC,

represented by $4 \times 0{\cdot}15 = 0{\cdot}6$

$$ac = \sqrt{0{\cdot}375^2 + 0{\cdot}6^2}$$

$$7 \times r = 0{\cdot}7075$$

$$r = 0{\cdot}101 \text{ m} = 101 \text{ mm}$$

$$\tan \alpha = \frac{0{\cdot}375}{0{\cdot}6} = 0{\cdot}625$$

$$\alpha = 32°$$

$$\theta = 32° + 90° = 122°$$

Therefore the 7 kg counter-balance should be placed at a radius of 101 mm, at 122° to the 3 kg mass. Ans.

Again note carefully that the actual forces are centrifugal forces, each of value $m\omega^2 r$, but since all rotate at the same angular velocity, each force can be *represented* by $m \times r$.

STRESS IN FLYWHEEL RIMS DUE TO CENTRIFUGAL FORCE

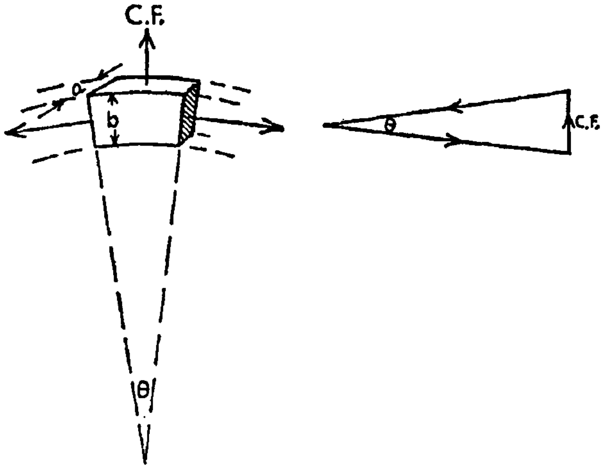

Fig. 73

Stress is the load or force carried by a material per unit of cross-section (see Chapter 9).

$$\text{Stress} = \frac{\text{total load}}{\text{area of cross section}}$$

Referring to Fig. 73, considering the equilibrium of a small piece of the flywheel rim it can be seen from the vector diagram of forces that the outward radial centrifugal force is balanced by the circumferential tensile force in the rim. This tension tends to snap the material, the stress, expressed by dividing the total tensile force by the area, is therefore termed the tensile stress or hoop stress.

length of piece $= r\theta$ $\quad (r = \text{mean rim radius})$

area of cross-section $= ab$

Mass of piece $= \text{volume} \times \text{density}$

$= r\theta \times ab \times \rho$

$C.F.$ of piece $= m\,\omega^2\,r$

$= r\theta \times ab \times \rho \times \omega^2 \times r$

From vector diagram,

$$\text{Tensile force} = \frac{C.F.}{\theta}$$

$$\text{Tensile stress} = \frac{\text{tensile force}}{\text{area of cross-section}}$$

$$= \frac{r\theta \times ab \times \rho \times \omega^2 \times r}{\theta \times ab}$$

$$= \rho\omega^2 r^2 = \rho v^2 \text{ N/m}^2$$

Example. Calculate the stress set up in a thin flywheel rim, 1 m mean diameter, made of steel of density 7860 kg/m^3, when rotating at 1500 rev/min.

$$\frac{150 \text{ rev/min} \times 2\pi}{60} = 157{\cdot}1 \text{ rad/s}$$

$$\text{Stress due to centrifugal force} = \rho v^2 = \rho\omega^2 r^2$$

$$= 7860 \times 157{\cdot}1^2 \times 0{\cdot}5^2$$

$$= 4{\cdot}85 \times 107 \text{ N/m}^2$$

$$= 48{\cdot}5 \text{ MN/m}^2$$

CONICAL PENDULUM

If a mass is suspended at the end of a cord and set in motion in a circular path in a horizontal plane, the system sweeps out the shape of a cone and therefore it is referred to as a 'conical pendulum' to distinguish it from the simple pendulum which swings backwards and forwards.

Referring to Fig. 74, if the mass travels at a constant angular velocity, the angle between the cord and the vertical centre-line remains constant because the system is in equilibrium, that is, the moment of the centrifugal force which tends to increase this angle is balanced by the moment of the weight of the mass which tends to close the angle (see Chapter 7).

The moment of a force about a point is the product of the force and its effective leverage. The effective leverage is the perpendicular distance from the line of action of the force to the point about which moments are taken. The centrifugal force acts radially outwards from the centre through the centre of gravity of the moving mass, this line of action is horizontal and, if taking moments about the point of suspension *o*, the perpendicular distance from the line of action of the centrifugal force to *o* is the

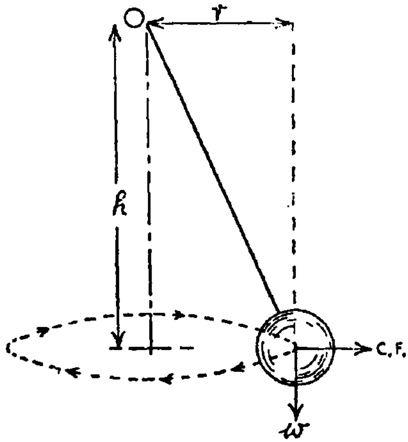

Fig. 74

vertical height h. The weight of the mass acts vertically downwards through its centre of gravity, the perpendicular distance from the line of action of this force to point o is the radius of the circular path r, therefore:

Let θ = angle between cord and vertical centre-line, taking moments about point o:

$$\text{Moment tending to increase } \theta = \text{moment tending to reduce } \theta$$
$$\text{centrifugal force} \times \text{h} = \text{weight} \times \text{r}$$

hence,

$$m\,\omega^2 r \times h = mg \times r$$

$$h = \frac{m \times g \times r}{m \times \omega^2 \times r}$$

$$h = \frac{g}{\omega^2}$$

Note that the height is inversely proportional to the square of the velocity and is independent of the mass.

SIMPLE UNLOADED GOVERNOR

The simple unloaded governor, often referred to as the Watt governor, works on the principle of the conical pendulum. This is illustrated in Fig. 75 and consists of a pair of bob masses suspended on arms from a centre spindle so that an increase in speed will cause the bobs to swing outward and upward; by connecting

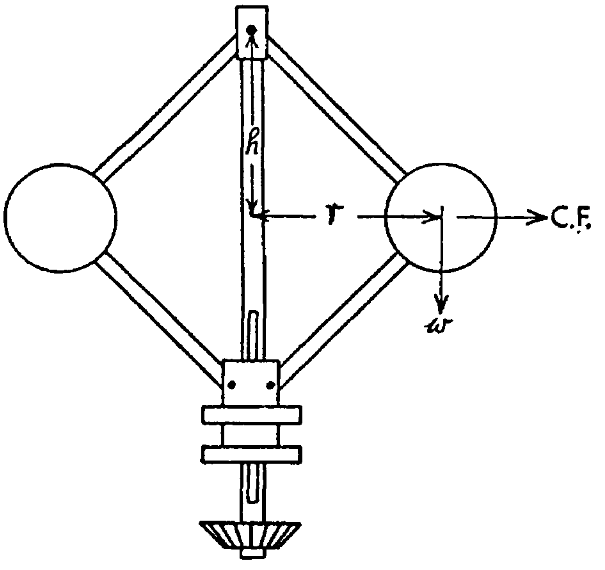

Fig. 75

through a pair of links to a sliding sleeve on the spindle, the sleeve is pulled up when the speed increases, and lowered when the speed decreases. This motion can be transmitted to an engine control valve by levers to close or open the valve, thus governing the engine speed within pre-determined limits. The height h from the plane of rotation of the bobs to the point of suspension, is referred to as 'the height of the governor'.

Example. Find the height of a Watt governor when rotating at speeds of 50 and 75 rev/min.

$$\frac{50 \text{ rev/min} \times 2\pi}{60} = 5{\cdot}236 \text{ rad/s}$$

$$h = \frac{g}{\omega^2} = \frac{9{\cdot}81}{5{\cdot}236^2}$$

$$= 0{\cdot}3579 \text{ m or } 357{\cdot}9 \text{ mm} \quad \text{Ans. (i)}$$

$$\frac{75 \text{ rev/min} \times 2\pi}{60} = 7{\cdot}854 \text{ rad/s}$$

$$h = \frac{g}{\omega^2} = \frac{9{\cdot}81}{7{\cdot}854^2}$$

$$= 0{\cdot}1591 \text{ m or } 159{\cdot}1 \text{ mm} \quad \text{Ans. (ii)}$$

Note. h is inversely proportional to ω^2

f PORTER GOVERNOR

The Porter governor is similar in its operation to the Watt governor but carries a central load on the sliding sleeve. The mass of this central load in relation to the mass of each bob determines the amount of movement of the sleeve for a given change of speed, and can be designed to suit engine control requirements.

Referring to Fig. 76 and considering the equilibrium of one bob. Let the links be all of equal length and take moments about the *instantaneous centre* (i) of the movement of the central load and the bob so that the effect of the weight of the central load is readily taken into account. h is the height from the plane of rotation of the bobs to point of suspension, and r is the radius of the circular path, therefore, since the links are of the same length, the height from the connection of the lower links and sleeve to plane of rotation is equal to h, and the distance from this connection to the instantaneous centre is $2r$ (by similar triangles).

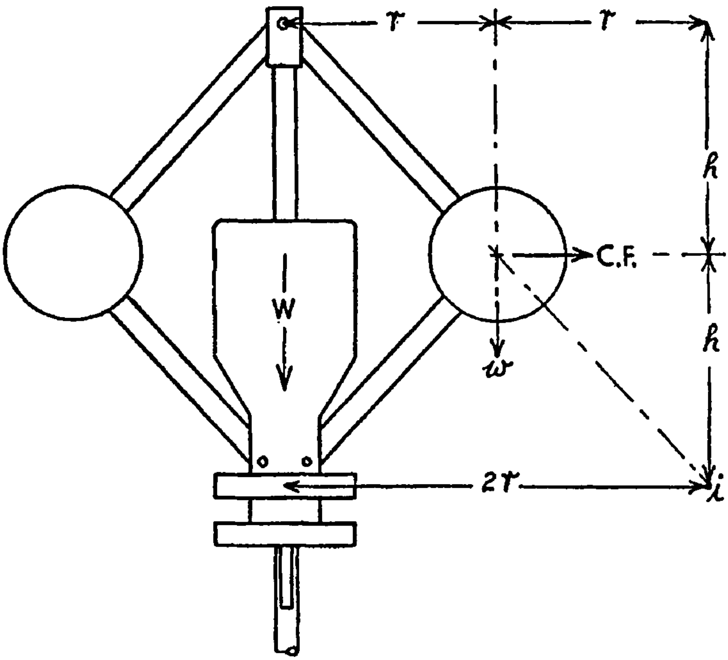

Fig. 76

Let M represent the mass of the central load and sleeve; m represents the mass of each bob.

Taking moments about i:—

$$\text{Clockwise moments} = \text{Anticlockwise moments}$$
$$C.F. \times h = mg \times r + \tfrac{1}{2}Mg \times 2r$$

Note that half the weight of the central mass is taken as being the effect on one bob,

$$C.F. \times h = gr\,(m + M)$$
$$h = \frac{gr\,(m + M)}{m\,\omega^2\,r}$$
$$h = \frac{g}{\omega^2}\left\{\frac{m + M}{m}\right\}$$

f Example. The links of a Porter governor are each 230 mm long. The mass of the central load and sleeve is 14 kg and the mass of each bob is 2 kg. When the governor rotates at a maximum speed the links make an angle of 60° to the vertical. Calculate (i) the maximum speed and, (ii) the change in height when the speed falls to 200 rev/min.

$$h = 230 \times \cos 60° = 115 \text{ mm} = 0{\cdot}115 \text{ m}$$
$$h = \frac{g}{\omega^2}\left\{\frac{m + M}{m}\right\}$$
$$0{\cdot}115 = \frac{9{\cdot}81}{\omega^2}\left\{\frac{2 + 14}{2}\right\}$$
$$\omega = \sqrt{\frac{9{\cdot}81 \times 8}{0{\cdot}115}} = 26{\cdot}12 \text{ rad/s}$$
$$\frac{26{\cdot}12 \text{ rad/s} \times 60}{2\pi} = 249{\cdot}4 \text{ rev/min} \qquad \text{Ans. (i)}$$
$$\text{height at 200 rev/min} = 115 \times \left\{\frac{249{\cdot}4}{200}\right\}^2$$
$$= 178{\cdot}8 \text{ mm}$$
$$\text{Change in height} = 178{\cdot}8 - 115$$
$$= 63{\cdot}8 \text{ mm} \qquad \text{Ans. (ii)}$$

EFFECT OF FRICTION. All frictional resistances in a governor, such as between the sleeve and spindle, at the various joints and in the operating gear can be reduced to one single friction force F acting at the sleeve. Friction always opposes motion, therefore when the speed of the governor is increasing and the sleeve rising, the friction force will act downwards, thus being equivalent to increasing the central load. When the speed of the governor is decreasing and the sleeve is descending, the friction force will act upwards and have the effect of reducing the centre load.

Hence, referring again to Fig. 76, considering the equilibrium of one bob, taking moments about i, let $+F$ represent the friction force at the central sleeve when the governor is increasing speed, and $-F$ when the governor speed is decreasing then:

$$C.F. \times h = mg \times r \ \tfrac{1}{2}(Mg \pm F) \times 2r$$

$$h = \frac{r(mg + Mg \pm F)}{m\,\omega^2 r}$$

$$h = \frac{g}{\omega^2}\left\{\frac{m + M \pm F/g}{m}\right\}$$

f Example. The mass of each bob of a Porter governor is 2 kg and the mass of the central load and sleeve is 19·2 kg. If the friction force at the sleeve is 7·85 N find the maximum and minimum governor speeds when the height is 250 mm.

$$h = \frac{g}{\omega^2}\left\{\frac{m + M \pm F/g}{m}\right\}$$

For maximum speed,

$$0{\cdot}25 = \frac{9{\cdot}81}{\omega^2}\left\{\frac{2 + 19{\cdot}2 - 7{\cdot}85/9{\cdot}81}{2}\right\}$$

$$\omega = \sqrt{\frac{9{\cdot}81 \times 22}{0{\cdot}25 \times 2}} = 20{\cdot}78 \text{ rad/s}$$

$$\frac{20{\cdot}78 \times 60}{2\pi} = 198{\cdot}4 \text{ rev/min} \qquad \text{Ans. (i)}$$

For minimum speed,

$$0{\cdot}25 = \frac{9{\cdot}81}{\omega^2}\left\{\frac{2 + 19{\cdot}2 - 7{\cdot}85/9{\cdot}81}{2}\right\}$$

$$\omega = \sqrt{\frac{9{\cdot}81 \times 20{\cdot}4}{0{\cdot}25 \times 2}} = 20 \text{ rad/s}$$

$$\frac{20 \times 60}{2\pi} = 191 \text{ rev/min} \qquad \text{Ans. (ii)}$$

f HARTNELL GOVERNOR

The Hartnell governor is a type of spring-loaded governor which is operated by a pair of bell-crank levers, the vertical arm of each lever carries a bob at its top end, the horizontal arm engages with the sleeve, and a central spring in compression exerts a controlling force on the sleeve, as diagrammatically shown in Fig. 77.

The force of gravity on each bob, *i.e.* its weight, has no effect on the sleeve when the arm carrying the bob is purely vertical. These arms lie slightly inwards at minimum speed, move outwards due to centrifugal force until they lie slightly outward from the vertical at maximum speed.

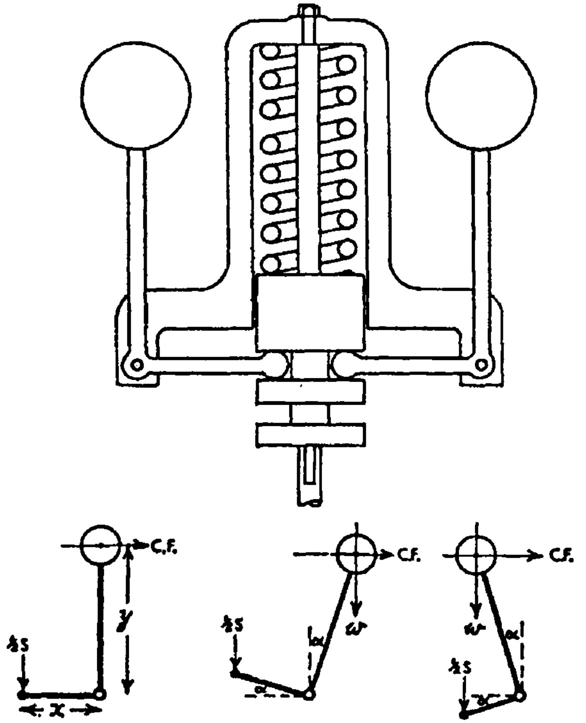

Fig. 77

Considering the equilibrium of each bob,

Let m = mass of each bob
w = weight of each bob = mg
r = radius of rotation of bobs
S = force exerted by spring on sleeve
y = length of vertical arm
x = length of horizontal arm

Taking moments about crank fulcrum,

At mean position:

$$C.F \times y = \tfrac{1}{2}S \times x$$

At high speeds:

$$C.F. \times y \cos\alpha + w \times y \sin\alpha = \tfrac{1}{2}S \times x \cos\alpha$$

At low speeds:

$$C.F. \times y \cos\alpha = w \times y \sin\alpha + \tfrac{1}{2}S \times x \cos\alpha$$

In most cases the angularity of the arms from the vertical and horizontal is small and therefore the moment of the weight of the bob about the crank fulcrum can usually be neglected.

If the weight of the sleeve is to be taken into consideration, it would be added to the spring force.

If the force to overcome sliding friction between the sleeve and the spindle is to be taken into account, it would be added to the spring force when the speed is increasing, and subtracted when the speed is decreasing, as previously shown.

f Example. In a Hartnell governor the mass of each bob is 1·5 kg. From the bell-crank lever fulcrum the length of the vertical arms to the centre of the bob is 120 mm and the length of the horizontal arms is 60 mm. When running at 300 rev/min the radius of rotation of the bobs is 80 mm, and at 320 rev.min the radius is 115 mm. Find the stiffness of the spring (N/mm) of compresssion. Neglect the effect of angularity of the arms from their respective vertical and horizontal positions.

$$\frac{300 \text{ rev/min} \times 2\pi}{60} = 31{\cdot}42 \text{ rad/s}$$

$$\frac{320 \text{ rev/min} \times 2\pi}{60} = 33{\cdot}52 \text{ rad/s}$$

Let S = force of spring on sleeve

$\tfrac{1}{2}S$ = downward force on end of each horizontal arm.

Considering equilibrium of each bob, taking moments about crank fulcrum,

At 300 rev/min:

$$C.F. \times y = \tfrac{1}{2}S_1 \times x$$

$$1{\cdot}5 \times 31{\cdot}42^2 \times 0{\cdot}08 \times 0{\cdot}12 = \tfrac{1}{2}S_1 \times 0{\cdot}06$$

$$S_1 = 473{\cdot}9 \text{ N}$$

At 320 rev/min:

$$1 \cdot 5 \times 33 \cdot 52^2 \times 0 \cdot 115 \times 0 \cdot 12 = \tfrac{1}{2} S_2 \times 0 \cdot 06$$

$$S_2 = 775 \cdot 2 \text{ N}$$

$$\text{Increase in compression} = 775 \cdot 2 - 473 \cdot 9$$

$$= 301 \cdot 3 \text{ N}$$

Horizontal movement at top of vertical arm

= difference in radius of rotation

$= 115 - 80 = 35$ mm

Ratio of lengths of vertical arm to horizontal arm

$= 120 \div 60 = 2$

Vertical movement at end of horizontal arm

$= 35 \div 2 = 17 \cdot 5$ mm this is the distance the sleeve moves up and therefore the amount the spring is compressed.

$$\text{Stiffness of spring} = \frac{301 \cdot 3}{17 \cdot 5}$$

$$= 17 \cdot 22 \text{ N/mm} \qquad \text{Ans.}$$

f SIMPLE HARMONIC MOTION

Simple harmonic motion is a particular form of reciprocating or 'to and 'fro' motion in which the acceleration and the velocity of the body varies as it moves from one end of its travel to the other, the important characteristic of this motion is that *the acceleration is proportional to the displacement from mid-travel (and directed to it).*

Note how this differs from anything which has been dealt with so far, previous to this, cases of *uniform* acceleration only have been explained.

When a body moves forwards and backwards with simple harmonic motion, at one end of its oscillation and the body is momentarily at rest, it receives maximum acceleration and causes it to move with increasing velocity. The acceleration, which is proportional to the displacement from mid-travel, decreases, and its velocity increases as the body approaches mid-travel. As it passes its mid-position the acceleration is zero and the velocity is maximum. From mid-travel onwards the acceleration is negative and increasing in magnitude while the velocity decreases until, at the other end of its travel, the velocity is momentarily nil and the acceleration is maximum to return the body in the opposite direction.

If a particle is travelling around a circular path in a vertical plane at a constant speed and its shadow could be seen on a horizontal table, the shadow would move forwards and backwards with simple harmonic motion.

From a practical point of view, neglecting the effect of the angularity of the connecting rod of a reciprocating engine, the piston moves with simple harmonic motion when the crank pin travels at a constant angular velocity.

Referring to Fig. 78, the circle represents the circular path of a particle Q moving at a constant velocity, this could be the crank pin of an engine. The projection of Q on to the plane of the diameter is point P, this could represent the relative position and motion of the piston (neglecting angularity of the connecting rod).

Q is moving at a constant angular velocity of ω around the circular path of r radius, its linear velocity is therefore ωr. The velocity vector diagram is drawn to represent the linear velocity of Q when passing the point at $\theta°$ past dead centre, the horizontal component represents the velocity of P at that instant. Thus:

$$\text{Velocity of P} = \text{Velocity of Q} \times \sin\theta$$
$$= \omega r \sin\theta$$

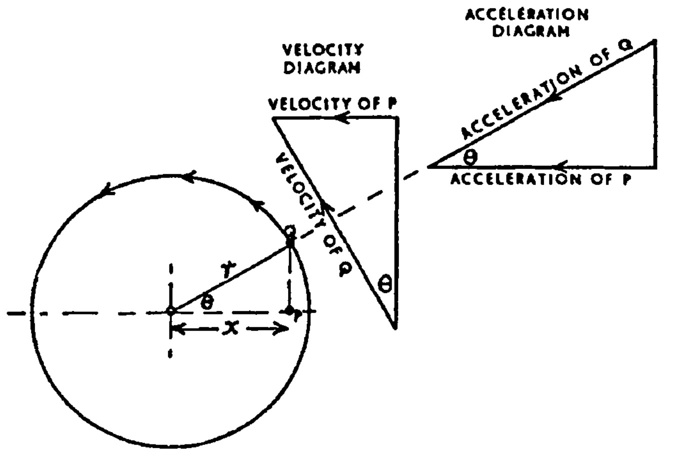

Fig. 78

If x represents the displacement of P from the middle of its travel, sin θ can be expressed in terms of r and x:

$$\text{Velocity of P} = \omega r \sin\theta$$
$$= \omega r \times \frac{\sqrt{r^2 - x^2}}{r}$$
$$= \omega\sqrt{r^2 - x^2}$$

The acceleration of a body moving around a circular path at constant velocity is centripetal acceleration, it has been shown to be $\omega^2 r$ and directed towards the centre. The acceleration of P is the horizontal component of the acceleration of Q, therefore:

$$\text{Acceleration of P} = \text{Acceleration of Q} \times \cos\theta$$
$$= \omega^2 r \cos\theta$$

or, expressing $\cos\theta$ in terms of x,

$$\text{Acceleration of P} = \omega^2 r \times \frac{x}{r}$$
$$= \omega^2 x$$
$$= \omega^2 \times \text{displacement from mid-travel}$$

The angular velocity ω is constant, therefore *the acceleration of a body moving with simple harmonic motion is proportional to its displacement from mid-travel (and directed to it).*

The AMPLITUDE is the maximum displacement to either side of its mid-travel.

The PERIODIC TIME is the time taken to make one complete oscillation, that is, to move completely across from one end to the other end and back again.

The time for P to make one complete oscillation is the same time that Q takes to move around one complete revolution, as follows,

$$\text{Time} = \frac{\text{distance}}{\text{velocity}}$$

$$\text{Time} = \frac{2\pi}{\omega}$$

The value of ω can be substituted into terms of displacement and acceleration.:–

$$\text{Acceleration} = \omega^2 \times \text{displacement}$$

$$\omega = \sqrt{\frac{\text{acceleration}}{\text{displacement}}}$$

$$\text{Periodic time} = \frac{2\pi}{\omega}$$

$$= 2\pi \div \sqrt{\frac{\text{acceleration}}{\text{displacement}}}$$

$$= 2\pi \sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

The above expression gives the periodic time for any body moving with simple harmonic motion.

The FREQUENCY is the number of oscillations made in 1 s.

If t = periodic time (s)

then, Frequency $= \frac{1}{t}$ osc/s

A MATHEMATICAL RELATION is given by:

$$x = r \cos \theta$$

$$x = r \cos wt \quad \textit{i.e.} \text{ displacement}$$

$$\frac{dx}{dt} = -\omega r \sin wt \quad \textit{i.e.} \text{ velocity } v$$

$$\frac{d^2x}{dt^2} = -\omega^2 r \cos wt \quad \textit{i.e.} \text{ acceleration } a \left(= \frac{dv}{dt}\right)$$

$$\frac{d^2x}{dt^2} = -\omega^2 x$$

The minus sign (mathematically correct) is because the acceleration is in the opposite direction to that of positive x (*i.e.* increasing). SHM, such that acceleration is directly proportional to displacement from a fixed point and directed towards it. The constant of proportionality is ω^2.

*f*Example. The stroke of a reciprocating engine is 500 mm. Neglecting the effect of the angularity of the connecting rod, find the velocity and acceleration of the piston when the crank is 30° past top dead centre and the engine is running at 750 rev/min.

Angular velocity of crank

$$= \frac{750 \times 2\pi}{60} = 78{\cdot}54 \text{ rad/s}$$

Radius of crank pin circle = ½ stroke = 0·25 m

Velocity of piston $= \omega r \sin \theta$

$= 78{\cdot}54 \times 0{\cdot}25 \times \sin 30°$

$= 9{\cdot}817$ m/s Ans. (i)

Acceleration of piston $= \omega^2 r \cos\theta$
$= 78{\cdot}54^2 \times 0{\cdot}25 \times \cos 30°$
$= 1336 \text{ m/s}^2$ Ans. (ii)

*f*Example. A machine component of 2·25 kg mass moves with simple harmonic motion, the amplitude being 380 mm. If it makes 120 osc/min, find (i) the maximum accelerating force, (ii) the accelerating force when the displacement is 250 mm from mid-travel.

120 osc.min = 2 osc/s

$\therefore$ Periodic time of one oscillation $= \frac{1}{2}$ s

Maximum displacement = 380 mm = 0·38 m

$$t = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

$$\tfrac{1}{2} = 2\pi\sqrt{\frac{0{\cdot}38}{\text{acceleration}}}$$

$$(\tfrac{1}{2})^2 = \frac{2^2 \times \pi^2 \times 0{\cdot}38}{\text{acceleration}}$$

$$\text{Acceleration} = 2^2 \times 2^2 \times \pi^2 \times 0{\cdot}38$$
$$= 60{\cdot}01 \text{ m/s}^2$$

$$\text{Accelerating force} = \text{mass} \times \text{acceleration}$$
$$= 2{\cdot}25 \times 60{\cdot}01$$
$$= 135 \text{ N}$$ Ans. (i)

Acceleration is proportional to displacement, therefore when the displacement is 0·25 m:

$$\text{Accelerating force} = 135 \times \frac{0{\cdot}25}{0{\cdot}38}$$
$$= 88{\cdot}82 \text{ N}$$ Ans. (ii)

f THE SIMPLE PENDULUM

A simple pendulum consists of a heavy bob swinging forwards and backwards suspended by a light cord.

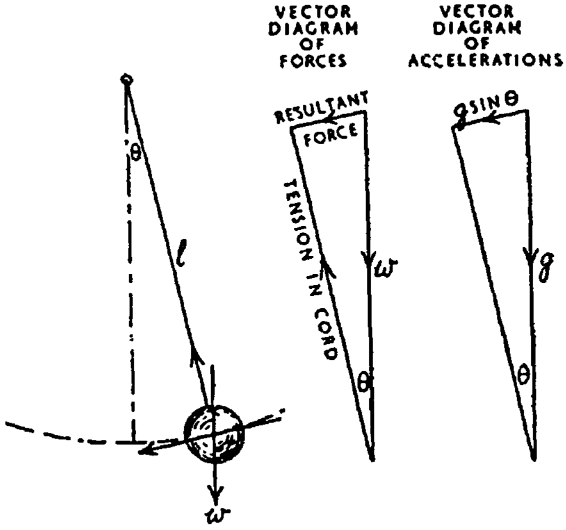

Fig. 79

Referring to Fig. 79, let m represent the mass of the bob, θ the angle made by the cord to the vertical centre line, and l the length of the cord. In the position shown, the effect of the force of gravity on the bob is to cause it to accelerate in the direction tangential to the arc of movement. The downward force of gravity on the bob is mg. From the vector diagram of the forces acting on the bob, the accelerating force causing it to move down the inclined plane is the component in that direction of the force of gravity, which is $mg \sin \theta$.

$$\text{Acceleration} = \frac{\text{accelerating force}}{\text{mass}}$$

$$= \frac{mg \sin \theta}{m} = g \sin \theta \quad \ldots \ldots \ldots \quad \text{(i)}$$

or this can be obtained direct from the vector diagram of accelerations, the acceleration down the plane being the component in that direction of the gravitational acceleration g.

The displacement of the bob from mid-travel of its swing is the length of the arc from the centre of swing to centre of bob,

$$\text{Displacement} = l\,\theta \quad \ldots \ldots \ldots \ldots \ldots \ldots \ldots \quad \text{(ii)}$$

From (i), acceleration is proportional to $\sin \theta$ because g is constant.

From (ii), displacement is proportional to θ radians because l is constant.

For small angles, sin θ equals θ rad, therefore for *small angles of swing*, acceleration is proportional to displacement and the motion is simple harmonic. Hence apply the expression for the periodic time of a body moving with simple harmonic motion:

$$\text{Periodic time} = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$

$$= 2\pi\sqrt{\frac{l\theta}{g \sin\theta}}$$

θ rad cancels with sin θ for a small angle, therefore,

$$t = 2\pi\sqrt{\frac{l}{g}}$$

Example. Find the length of a simple pendulum to make (i) one complete osc/s (*i.e.* one swing), (ii) half an osc/s.

In the first case the periodic time is 1 s,

$$t = 2\pi\sqrt{\frac{l}{g}}$$

$$l = \frac{t^2 \times g}{2^2 \times \pi^2}$$

$$= \frac{1 \times 9{\cdot}81}{2^2 \times \pi^2}$$

$$= 0{\cdot}2485 \text{ m}$$

$$= 248{\cdot}5 \text{ mm} \qquad \text{Ans. (i)}$$

In the second case, the periodic time for one complete oscillation is 2 s,

t varies as $\sqrt{l}$

$\therefore$ l varies as t^2

$\therefore l = 248{\cdot}5 \times (2/1)^2$

$= 994$ mm Ans. (ii)

f VIBRATIONS OF A SPRING

If a mass is attached to the end of a spring and the spring is allowed to extend gradually, an equilibrium condition will be reached when the spring force equals the weight of the attached mass.

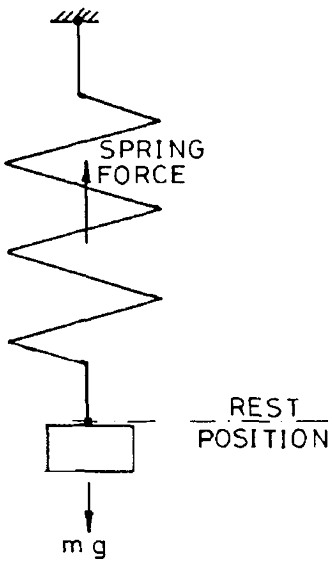

Fig. 80(a)

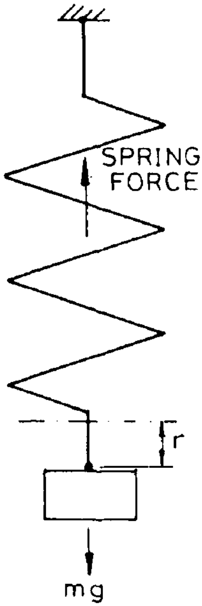

Fig. 80(b)

Thus, referring to Fig. 80(a)

With the spring at rest,

$$\text{Spring force} = mg$$

If the mass is now pulled down some distance r below the rest position and then released, the mass will oscillate vertically through a distance r above and below the rest position.

Referring to Fig. 80(b)

Let S = spring stiffness

Thus, when the spring has been extended a distance r

$$\text{Spring force} = mg + Sr$$

At the instant of release,

$$\text{Resultant upward force} = mg + Sr - mg$$
$$= Sr$$
$$\text{Force} = \text{mass} \times \text{acceleration}$$
$$Sr = m \times a$$
$$a = \frac{Sr}{m}$$

Since stiffness S and mass m are constants for a given arrangement, acceleration must be proportional to displacement r.

Hence, the mass oscillates vertically with simple harmonic motion.

$$\text{Periodic time } t = 2\pi\sqrt{\frac{\text{displacement}}{\text{acceleration}}}$$
$$t = 2\pi\sqrt{\frac{r}{Sr/m}}$$
$$\text{Periodic time} = 2\pi\sqrt{\frac{m}{S}}$$

Example. A close coiled helical spring has a stiffness of 1400 N/m. If a 10 kg mass is attached to the end of the spring, calculate the periodic time and the frequency of oscillation when the spring vibrates vertically.

$$\text{Periodic time } t = 2\pi\sqrt{\frac{m}{S}}$$
$$= 2\pi\sqrt{\frac{10}{1400}}$$

$$\text{Time for one oscillation} = 0{\cdot}531 \text{ s} \qquad \text{Ans. (i)}$$

$$\text{Frequency} = \frac{1}{\text{periodic time}}$$
$$= \frac{1}{0{\cdot}531}$$
$$\text{Frequency} = 1{\cdot}88 \text{ osc/s} \qquad \text{Ans. (ii)}$$

EFFECT OF THE MASS OF THE SPRING. If the mass of the spring is taken into account, it can be shown that its effect is equivalent to one-third of its mass being placed at the free end

$$\text{Periodic time } t = 2\pi\sqrt{\frac{m + \frac{m_s}{3}}{S}}$$

Example. A helical spring with a mass of 6 kg has a mass of 10 kg attached to its free end. The mass is now pulled down 40 mm from the equilibrium position and then released. (i) If the spring stiffness is 532 N/m find the frequency of oscillation. (ii) Determine also the kinetic energy of the mass when it is 30 mm from the equilibrium position.

(i)

$$\text{Periodic time} = 2\pi\sqrt{\frac{m + \frac{m_s}{3}}{S}}$$

$$= 2\pi\sqrt{\frac{10 + \frac{6}{3}}{532}}$$

$$= 0{\cdot}944 \text{ s}$$

$$\text{Frequency} = \frac{1}{0{\cdot}944}$$

$$= 1{\cdot}06 \text{ osc/s} \qquad \text{Ans. (i)}$$

$$\text{Periodic time } t = \frac{2\pi}{\omega}$$

$$\omega = \frac{2\pi}{0{\cdot}944}$$

$$= 6{\cdot}66 \text{ rad/s}$$

For this spring motion,

$$\text{Amplitude } r = 0{\cdot}04 \text{ m}$$

When displacement x from mid-travel is 0·03 m:

$$\text{Instantaneous velocity} = \omega\sqrt{r^2 - x^2}$$

$$= 6{\cdot}66\sqrt{0{\cdot}04^2 - 0{\cdot}03^2}$$

$$= 0{\cdot}176 \text{ m/s}$$

$$\text{Translational K.E.} = \tfrac{1}{2}mv^2$$

$$= \tfrac{1}{2} \times 10 \times 0{\cdot}176^2$$

$$= 0{\cdot}155 \text{ J.} \qquad \text{Ans. (ii)}$$

TEST EXAMPLES 5

1. A mass of 1·2 kg is connected to the end of a cord and rotated in a vertical plane, the radius of the circular path being 600 mm. Find (i) the maximum tension in the cord when the speed of rotation is 75 rev/min, (ii) the speed of rotation when the minimum tension is nil.

2. A hole is bored in a circular disc of uniform thickness. The centre of the hole is 38 mm from the centre of the disc and the mass of material removed is 0·8 kg. Calculate the centrifugal force set up when the bored disc rotates about its geometrical centre at 240 rev/min.

3. A motor car is on the verge of skidding when travelling at 48 km/h on a level road around a curvature of 30 m radius. Find the coefficient of friction between the tyres and the ground.

4. The wheel track of a motor van is 1·68 m and its centre of gravity when fully loaded is 0·98 m above the ground. Calculate the speed at which the van will overturn when travelling around a bend of 23 m radius, assuming the road to be level.

5. Calculate the super-elevation of the outer rail of a curved railroad of 250 m radius so that there will be no side thrust when a train is travelling at 75 km/h, the track being 1·435 m.

6. Three masses, x, y and z, are to rotate in the same plane at the same angular velocity. Their masses and radii from centre of rotation are, respectively 9 kg at 400 mm radius, 10 kg at 350 mm, and 12 kg at 250 mm. Calculate the angles between them so that they will balance.

7. A thin flywheel rim, 1·2 m diameter, is made of cast iron of density 7210 kg/m^3. Find the speed in rev/min at which the stress due to centrifugal force will be 15 MN/m^2.

8. Find the change in height of a simple unloaded governor when it changes speed from 60 to 80 rev/min.

*f*9. The two balls of a Porter governor each have a mass of 2·25 kg. Assuming all links to be the same length, find the mass of the central load so that the change in height will be 25 mm when the speed changes from 240 to 270 rev/min.

*f*10 A push-rod moves with simple harmonic motion driven by an eccentric sheave running at 90 rev/min, the full travel of the rod being 50 mm. Calculate its velocity and acceleration when it is 6 mm from the beginning of its travel.

*f*11. The effective force on the piston of a vertical diesel engine when passing top dead centre is 800 kN, the mass of the reciprocating parts is 1524 kg, the length of the stroke is 1100 mm, and the engine is running at 120 rev/min. Assuming the motion of the reciprocating parts to be simple harmonic, find the effective thrust on the crosshead at the beginning of the down stroke.

*f*12. A cam rotates at 3 rev/s and imparts a vertical lift of 60 mm to a follower of mass 2 kg. The follower moves with S.H.M. and the lift of the follower is completed in one-third of a revolution of the cam. Calculate (a)the maximum acceleration of the follower and (b)the maximum force between cam and follower.

*f*13. Calculate the length of a simple pendulum to make 120 complete oscillations per minute.

*f*14. The stiffness of a helical spring is such that it stretches 1 mm for every 1·5 N of load. Find the number of vibrations it will make per minute if it is disturbed from rest when carrying a mass of 6·5 kg.

*f*15. A helical spring has a stiffness of 25 kN/m.

If a mass of 100 kg is attached to its free end, pulled down, and then released, determine the periodic time of its motion.

If the maximum deflection was 50 mm, find the velocity and acceleration of the mass when it is 300 mm from the equilibrium position.